



2026

Fleet Worth



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Problem



Pricing a truck based on images and description that a user uploads for reselling requires human expertise and experience in reselling trucks.

The process is further complicated by the way a truck's value depends heavily on information the seller controls, like condition, mileage, visible damage, tires, configuration, etc. But a marketplace cannot simply assume that self-reported information is complete or accurate.



Process



1

**We wanted to remove
dependence on self-reported
condition.**

2

**Scoping out the right approach:
Image → Video →
Real time + Hybrid**



Data

Don't want to rely on VIN, mileage, license plate, or other identifiers.

1. We scraped 6 sites, over 67,563 truck listings using distributed data scraping pipeline
2. Cleaned dataset, created embedding, and built DINO index → 57,809 truck listings



Jerry

Solution ↙

- User takes live video directly in the app to prevent forging
- Frames are auto-snapped
- User receives guidance for what they should take aim camera
- Frames are analyzed and sent to our VLM, which returns a JSON of make, model, estimated year, and condition, as well as verifying the truck is real





Solution ↙

- DINOv2 model then encodes the image into a 768-dimensional normalized embedding, retrieves the nearest indexed listing by cosine similarity, and estimates the price from a weighted distribution of those comparable trucks
- This accounts for potential damages
- We combine the results from the JSON describing the truck and similar trucks using vector similarity with DINOv2 to produce the output



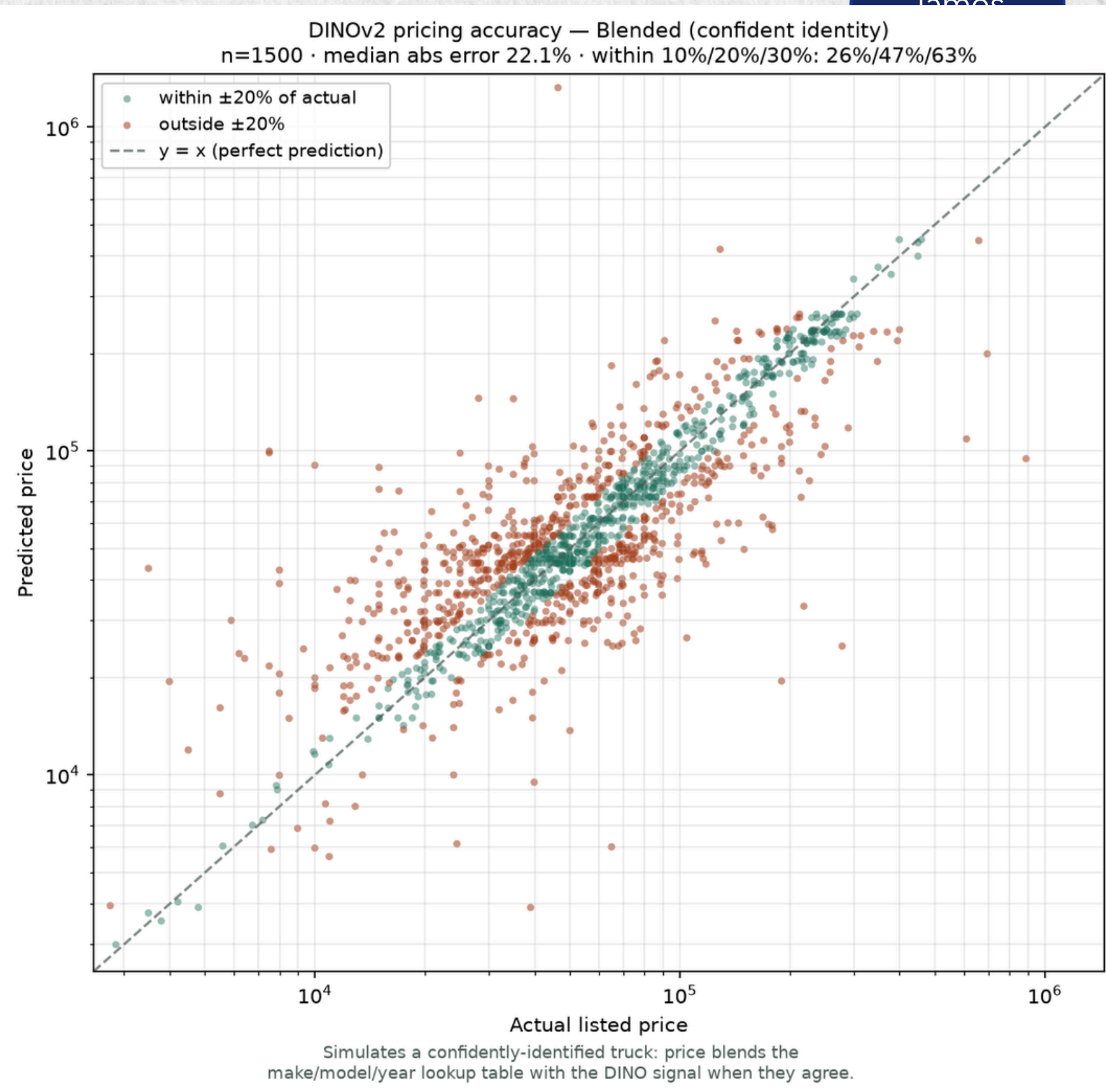
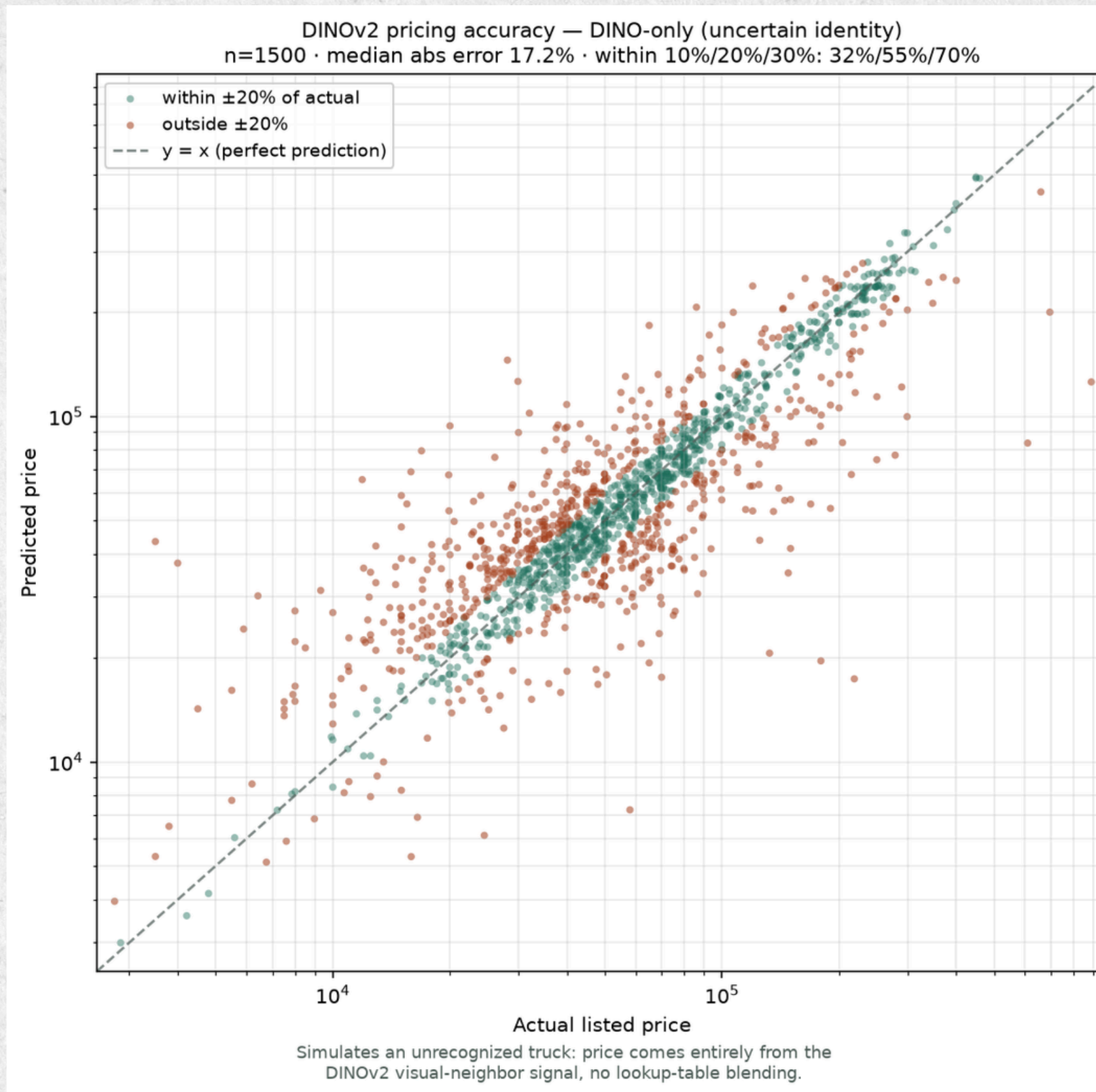
Stats

James

67,563 raw scraped lines	58,723 cleaned & deduped listings	57,809 DINOv2-embedded comps	4 independent machines
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PER-MACHINE EMBEDDED SHARD

MACHINE	ROLE	ITEMS EMBEDDED
runpod-1	RunPod pod · RTX 4090	57,376
macbook-pro	Local Mac (scrape + embed)	54,670
runpod-2	RunPod pod · RTX 5090	54,655
james-pc	Teammate Windows PC	45,972



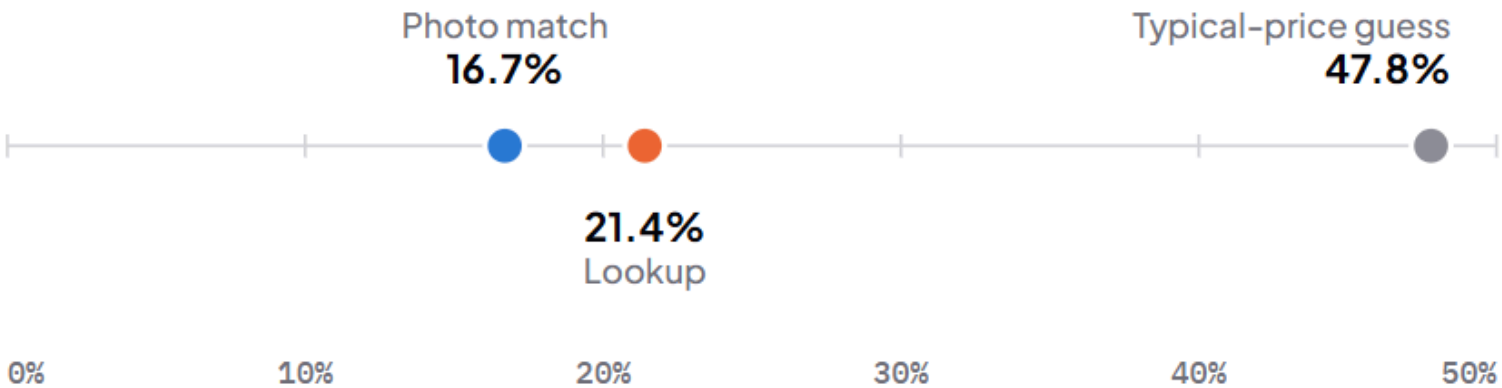


PRICING ACCURACY • 5,000 HELD-OUT TRUCKS

16.7%

typical miss between Fleetworth’s photo-based price and the dealer’s asking price.

MEDIAN MISS VS. ASKING PRICE



56.4% of trucks priced within $\pm 20\%$ of their asking price from photo matches alone

79.1% of quoted price ranges contain the actual asking price

1.003 median predicted $\div$ asking price: about half of trucks land below asking and half above

2.9 $\times$ smaller typical miss than quoting the dataset’s median truck price





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Demo

